

MATTHEW A. WRIGHT

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RESEARCH VISION

I am a solid-state materials chemist working on battery and quantum materials. I study how structure governs function in inorganic solids to develop rational design principles for new functional materials. I combine solid-state synthesis with electrochemistry as a synthetic tool for tuning composition and electronic structure and use X-ray and neutron scattering alongside spectroscopy to resolve the structural origins of ionic, electronic, and magnetic behavior.

EMPLOYMENT

- 2024– Postdoctoral associate, Materials Department, College of Engineering, UC Santa Barbara, California, USA.
Developing new chemical compounds and processes for batteries and quantum materials.
Advised by Professor Ram Seshadri.
- 2024 Honorary research assistant, Stephenson Institute for Renewable Energy, University of Liverpool, UK.
Exploring multi-valent ion insertion mechanisms in covalent framework materials.
Advised by Professor Laurence J. Hardwick.

EDUCATION

- 2019–2023 PhD in Solid-State Chemistry, Materials Innovation Factory, Department of Chemistry, University of Liverpool, UK.
Thesis titled *Structure and Properties of New Positive Electrode Materials for Magnesium-Ion Batteries*.
Advised by Professor Matthew J. Rosseinsky OBE FRS.
- 2015–2019 MChem (Hons) in Chemistry, Department of Chemistry, University of Liverpool, UK. Includes study abroad at Boston College, Massachusetts, USA.
Master's thesis titled *Discovery & Characterization of New Pb-free Piezoelectric-Relaxor Ceramic Materials*.
Advised by Professor Matthew J. Rosseinsky OBE FRS.
Bachelor's thesis titled *Covalent Polymers for the Catalysis of Hydrogen Evolution from Water*.
Advised by Professor Andy Cooper FRS.

PUBLICATIONS

20 peer-reviewed, 5 under review; 8 first-author, 4 corresponding author*.

PERSPECTIVES

25. MA Wright*, R Seshadri* "Linking Battery Electrode Science with Correlated and Quantum Materials" *Phys. Rev. Mater.* **9**, 110302 (2025).

ARTICLES

24. MA Wright*, EC Moya, S Browne, AJ Bologna, KE Brockmeyer, AR Reach, R Seshadri* "Na-ion Cathodes in Minutes: Microwave Synthesis Across the Na-Mn-Fe-Ni-Cu-O Composition Space" (*accepted & in press, ACS Appl. Energy Materials*).
23. T Islam, S Bayat, K Pitton, MA Wright, MM Rahman, R Yin, S Clark, SC Roy, R Feng, R Chernikov, KM Wiaderek, LC Gallington, AM Abeykoon, C Risko, R Amin, BS Guiton, MS Islam* "Covalently Integrated Amorphous $\text{Mn}_x\text{Mo}_3\text{S}_{13}$ -GO Chalcocarbogel for High Capacity and Long-Life Sulfur Equivalent Conversion Electrodes" (*Small*, *Early Access*).
22. EA Lawrence, MA Wright, T Li, EN Bassey, V Kumar, P Cardon, S Yuan, PE Cabelguen, RJ Clément* "Monitoring the Microwave Synthesis of d^0 -Free Disordered Rocksalt Cathodes Using *in Situ* Infrared Pyrometry" (*Angew. Chem. Int. Ed.*, *Early Access*).
21. MA Wright*, S Browne, EC Moya, AR Reach, A Sebastian, A Abulajiang, R Yin, J Cesareo, J Li, A Tarin, SH Tolbert, BC Melot, A Van der Ven, R Seshadri* "Rapid and Phase-Selective Routes to Layered Na-Ion Cathodes" *Chem. Mater.* **38**, 8476–8488 (2026).
20. A Morscher, L Corti, SL Goodwin, MA Wright, TW Surta, MS Dyer, F Blanc, LM Daniels, JB Claridge, MJ Rosseinsky* "High Lithium Content and Site Disorder in the Transition Metal Oxide Argyrodites $\text{Li}_7\text{TiO}_5\text{X}$ ($\text{X} = \text{Cl}^-$, Br^-)" *Chem. Mater.* **38**, 6312–6322 (2026).
19. A Reach, MA Wright, AS Mulligan, Z Fang, KT Tseng, C Wang, JG Hu, M Chi, AK Cheetham, R Seshadri, J Sakamoto* "Enhanced Sodium Dynamics in Biphasic NaSICON Solid Electrolytes" *Chem. Mater.* **38**, 5356–5368 (2026).
18. H Tian, T Li, AS Mulligan, J Tregidga, MA Wright, A Zohar, A Chezhiyan, MB Preefer, KH Stone, J Hu, G Wu, AM Manjón-Sanz, SH Lapidus, JW Harter, AK Cheetham*, R Seshadri* "Structural Properties in Cs_2MBiX_6 ($\text{M} = \text{Na}$, Ag ; $\text{X} = \text{Cl}$, Br) Bismuth Halide Double Perovskites" *Chem. Mater.* **38**, 3350–3360 (2026).

17. KE Brockmeyer, AJ Bologna, MA Wright, J Wong, C Rodriguez, T Li, RA Segalman, R Seshadri* "Direct Microwave Pyrolysis of Cellulose to Hard Carbon Anodes for Sodium-Ion Batteries" *Chem. Mater.* **38**, 1377–1384 (2026).
16. MA Wright, AS Mulligan, D Rout, JG Hu, RL Behrens, JR Chamorro, SD Wilson, AK Cheetham, R Seshadri* "Strong, Yet Split Hydrogen Bonding with Ice Rules in Delafossite (H/D)RhO₂" *Angew. Chem. Int. Ed.* **137**, e15471 (2025).
15. MA Wright, AR Neale, AA Lalanza, H Gao, MJ Rosseinsky, AI Cooper, LJ Hardwick* "Revealing the Mg-Ion Storage Mechanism Within a Covalent Organic Framework Electrode" *ACS Appl. Energy Mater.* **8**, 16451–16456 (2025).
14. MA Weret, T Islam, S Bayat, SC Roy, C Sawicki, P Powe, CL Donley, AS Kumbar, AM Abeykoon, R Chernikov, RM Curtis, MA Wright, KM Wiaderek, C Risko, R Amin, SM Islam* "Semiconducting Zn_xMo₃S₁₃-GO Chalcocarbogel: a High-Capacity and Stable Sulfur-Equivalent Conversion-Based Electrode for Lithium-Ion Batteries" *Chem. Mater.* **37**, 5466–5475 (2025).
13. A J Bologna, RC Vincent, A Kallistova, JA Mayer, MA Wright, CR Dela Cruz, R Zhang, F Seeler, K Schierle-Arndt, R Seshadri* "Pyrolyzed Black Mass Feedstocks and Their Synthetic Proxies Relevant to Li-Ion Battery Recycling" *ACS Omega* **10**, 25546–25554 (2025).
12. AR Reach, A Zohar, MA Wright, T Li, R Seshadri* "Rapid Microwave Preparation of AlNb₁₁O₂₉ and Ti₂Nb₁₀O₂₉ Wadsley–Roth Lithium-Ion Anode Compounds from Parent Oxides" *ACS Appl. Energy Mater.* **8**, 7714–7721 (2025).
11. T Islam, S Bayat, MA Wright, SC Roy, C Sawicki, CL Donley, AS Kumbar, R Chernikov, MA Weret, KM Wiaderek, C Risko, R Amin, S Islam* "Porous Amorphous Mn_xMo₃S₁₃ Chalcogel Electrode for High-Capacity Conversion-Based Lithium-Ion Batteries" *J. Am. Chem. Soc.* **147**, 7400–7410 (2025).
10. MA Wright, J Lim, RA Pacheco, AE Krowitz, CJ Hawkins, M Bahri, LM Daniels, R Chen, LG Chagas, J Cookson, P Collier, AV Chadwick, ND Browning, JB Claridge, LJ Hardwick*, MJ Rosseinsky* "Fast Mg-ion insertion kinetics in V₂Se₉" *J. Mater. Chem. A* **12**, 32349–32358 (2024).
9. N Hulai, M Zanella, C Robertson, D Ritchie, M Sonni, MA Wright, JA Newnham, CJ Hawkins, J Whitworth, B Mali, H Niu, M Dyer, CM Collins, LM Daniels, JB Claridge, MJ Rosseinsky* "Navigation Through High-Dimensional Chemical Space: Discovery of Ba₅Y₁₃[SiO₄]₈O_{8.5} and Ba₃Y₂[Si₂O₇]₂" *Chem. Sci.* **15**, 16503–16518 (2024).
8. MA Wright, TW Surta, JA Evans, J Lim, H Jo, CJ Hawkins, M Bahri, LM Daniels, R Chen, M Zanella, LG Chagas, J Cookson, P Collier, G Cibir, AV Chadwick, MS Dyer, ND Browning, JB Claridge, LJ Hardwick*, MJ Rosseinsky* "Accessing Mg-Ion Storage in V₂PS₁₀ Via Combined Cationic–Anionic Redox with Selective Bond Cleavage" *Angew. Chem. Int. Ed.* **63**, e202400837 (2024).
7. CJ Hargreaves, MG Gaultois, LM Daniels, EJ Watts, VA Kurlin, M Moran, Y Dung, R Morris, A Morscher, K Thompson, MA Wright, BE Prasad, F Blanc, CM Collins, CA Crawford, BB Duff, J Evans, J Gamon, G Han, BT Leube, H Niu, AJ Perez, A Robinson, O Rogan, PM Sharp, E Shoko, M Sonni, WJ Thomas, A Vasylenko, L Wang, MJ Rosseinsky*, MS Dyer* "A Database of Experimentally Measured Lithium Solid Electrolyte Conductivities Evaluated with Machine Learning" *NPJ Comp. Mater.* **9**, 1935 (2023).
6. TW Surta, TA Whittle, MA Wright, H Niu, J Gamon, QD Gibson, LM Daniels, WJ Thomas, M Zanella, PM Shepley, Y Li, A Goetzee-Barral, AJ Bell, J Alaria, JB Claridge, MJ Rosseinsky* "One Site, Two Cations, Three Environments: s² and s⁰ Electronic Configurations Generate Pb-Free Relaxor Behaviour in a Perovskite Oxide" *J. Am. Chem. Soc.* **143**, 1386–1398 (2021).

UNDER REVIEW

5. MA Wright*, S Browne, J Mayer, D Rout, SD Wilson, AK Cheetham, A Van der Ven, R Seshadri* "Vacancy Ordering and Correlated Electron Effects in Layered Na_{1-x}RhO₂" (*under review*).
4. Q Zhang, CH Shiao, D Graf, MA Wright, X Zhu, D Davis, Y Zhao* "Polyol Synthesis of Water-Dispersible Hard-Soft CoFe₂O₄@ZnFe₂O₄ Core-Shell Nanoparticles for Enhanced MRI and MPI" (*under review*).
3. P Zarabadi-Poor, J Lim, X Wang, JD Bocarsly, MA Wright, AA Lalanza, LM Daniels, IM Pateli, AR Neale, R Chen, JB Claridge, CP Grey, BJ Morgan, LJ Hardwick*, MJ Rosseinsky*, MS Islam* "Accessing Polyanionic Redox in High Voltage Li-Rich Thiophosphates" (*under review; Archive*).
2. MA Weret, A Adebajo, T Islam, K Pitton, T Islam, CL Donley, AS Kumbhar, BS Guiton, AM Abeykoon, MA Wright, SC Roy, O Oketola, R Amin, SM Islam* "Short-Range-Ordered Mo₃S₁₃-Graphene Oxide Chalcocarbogel Electrodes for High-Capacity and Cycle-Stable Lithium-Ion Batteries" (*under review*).
1. A Malik, A Gardner, CJ Hawkins, D Inns, H Jang, M Sonni, MA Wright, M Bahri, M Alfredson, AV Chadwick, T Manning, R Moca, I Casely, G Stevens, J Stevens, R Chen, AJ Cowan, MJ Rosseinsky* "Ir–Sb–Sn–O Solid Solutions as Acid-Stable OER Catalysts: Structural Origin of Improved Ir Utilization Compared to Supported IrO_x" (*under review*).

INVITED TALKS

Invited Speaker & Poster Judge, Partnership for Research and Education in Materials, 2025 – JSU, Mississippi, USA.

Talk titled "Protocols for Microwave-Assisted Synthesis of Inorganic Oxides".

Invited Speaker, Johnson Matthey Academic Conference, 2023 – University of Warwick, Coventry, UK.

Talk titled "Unlocking Combined Cationic-Anionic Multi-Electron Redox in Mg Batteries" (best oral presentation prize).

SELECT PRESENTATIONS

- Gordon Research Conference on Solid State Chemistry, 2026 – Colby-Sawyer College, New Hampshire, USA
Poster titled “Strong, yet split hydrogen bonding with ice rules in delafossite (H/D)RhO₂”.
- Artificial Intelligence Science at Scale Annual Meeting, 2026 – Santa Fe, New Mexico, USA
Co-presented a poster titled “New Materials for Topological Switches”.
- Materials Research Outreach Program, 2026 – Santa Barbara, California, USA
Poster titled “Strong, yet split hydrogen bonding with ice rules in delafossite (H/D)RhO₂”.
- American Chemical Society Fall Meeting, 2025 – Washington, District of Columbia, USA.
1st talk titled “Understanding hydrogen bonding and residual entropy in delafossite structures”.
2nd talk titled “Accessing Mg-Ion Storage via Combined Cationic-Anionic Redox with Selective Bond Cleavage”.
- 16th International Workshop on Advanced Materials, 2025 – Ras Al Khaimah Center for Advanced Materials, RAK, UAE.
Poster titled “Electrochemical Control of Quantum Materials Response”.
- Materials Research Society Fall Meeting, 2024 – Boston, Massachusetts, USA.
Talk titled “Accessing Mg-Ion Storage via Combined Cationic-Anionic Redox with Selective Bond Cleavage”.
- Neutron Scattering Workshop, 2024 – Oak Ridge National Laboratory, Oak Ridge, Tennessee, USA.
Poster titled “Electrochemical Control of Quantum Materials Response”.
- 241st Electrochemical Society Meeting, 2022 – Vancouver, British Columbia, Canada.
Poster titled “Anionic Redox and Multi-Electron processes in Mg Batteries”.
- Faraday Summer Internship Conference, 2020 – University of Warwick, Coventry, UK.
Poster titled “Computationally Predicted Doping of Aluminosilicates for Li-Ion Conductivity” (runner-up poster prize).
- Undergraduate Research Symposium, 2019 – University of Liverpool, Merseyside, UK.
Talk titled “Discovery and Characterization of New Pb-free Perovskite Relaxors for Piezoelectric Applications”.

TECHNICAL SKILLS**BATTERY DEVELOPMENT & ELECTROCHEMISTRY**

Li-, Na-, and Mg-ion systems; electrode formulation and casting; cell assembly; galvanostatic cycling, cyclic voltammetry, and GITT/PITT diffusion measurements; design and execution of *operando* experiments.

MATERIALS SYNTHESIS & PROCESSING

Solid-state synthesis of oxides and chalcogenides; flux and vapor-transport crystal growth; ceramic processing; air-sensitive handling (glovebox, Schlenk line); fused-silica and metal ampoules.

CHARACTERIZATION & ANALYSIS

Bragg and diffuse X-ray and neutron scattering, including Rietveld refinement and Fourier / maximum-entropy analysis; XPS and HAXPES, XANES, solid-state NMR, Raman and IR, XRF, ICP-OES; SQUID magnetometry and PPMS transport measurements.

SOFTWARE, DATA & LAB OPERATIONS

Python and JupyterLab for data analysis; TOPAS, GSAS-II, Jana, PDFgui, VESTA (crystallography); CasaXPS, Demeter, TopSpin, EC-Lab, ZView; LaTeX, OriginPro, GIMP. Laboratory safety officer; glovebox maintenance; cryogen and gas handling; fire and first-aid trained.

AWARDS AND SCHOLARSHIPS

- Heeger Fellowship, California NanoSystems Institute, UC Santa Barbara — \$1,500 awarded, 2026.
- Future Leaders in Advanced Materials travel grant, Materials Research Lab, UC Santa Barbara — \$800 awarded, 2025.
- Winning Image, Materials Department scientific image competition, UC Santa Barbara, 2025.
- Johnson Matthey Academic Conference best oral presentation award, 2023.
- Summer Internship Conference runner-up poster prize, 2020.
- Faraday Undergraduate Summer Intern — £5,800 stipend awarded, 2019.
- Santander Global Mobility Award — £1,000 supporting an exchange year at Boston College, USA, 2016–2017.

TEACHING & MENTORING

- 2026 Guest lecturer for UCSB Materials 218/Chemistry 277: Delivered a lecture on crystal structures and bonding in inorganic solids.
- 2025– Mentored student interns via the NSF-supported Partnership for Research in Energy Materials and Future Leaders in Advanced Materials programs. Provided research mentorship and training to undergraduate researchers from underrepresented groups at partner institutions (JSU/UTEP), supporting their development in materials synthesis, characterization techniques, and scientific communication.

- 2023– Extensive mentoring of graduate and undergraduate researchers (with current location):
 [UC Santa Barbara]: Amelia Reach (University of Oxford, PhD), Enrique Moya (Helmholtz Institute Ulm, PhD), Roland Yin (Columbia University, PhD), Alejandro Tarin (University of Texas, El Paso, PhD), Presley Cochran (Leeta Materials).
 [University of Liverpool]: Clément Morel (Collège de France, PhD), Raul Pacheco Muino (Imperial College London, PhD).
- 2019–2023 Graduate teaching assistant, Department of Chemistry, University of Liverpool. [Weblink](#) to course content.
 Solid State Chemistry (Chem442, Level 7 FHEQ)
 Third- and Fourth-Year Research Project (Chem370/380, Chem480, Level 6–7 FHEQ)
 Applications for Group Theory (Chem216, Level 5 FHEQ)
- 2018–2022 Private tutor, TutorRight. Teaching mathematics, chemistry, and physics (UK Key Stage 2 through A-Level).
- 2016–2018 Leader for Peer-assisted learning (PAL) (University of Liverpool): a student-driven scheme organizing and facilitating weekly study support sessions for new undergraduates.

PROFESSIONAL DEVELOPMENT

- Rice Centre for Quantum Materials, Workshop on Quantum Geometry – Rice University, TX, USA, November 2025.
- Rice Centre for Quantum Materials, School on Quantum Materials Synthesis – Rice University, TX, USA, November 2025.
- Advanced Neutron Powder Diffraction Analysis with TOPAS – Oak Ridge National Laboratory, TN, USA, September 2024
- Leadership in Academia Workshop, Professor Matthew Reed – University of Liverpool, UK, July 2023
- EPSRC Doctoral Training Centre in Digital & Automated Materials Chemistry – University of Liverpool, UK, 2019–2023
- Faraday Institute Summer Internship – Faraday Institute, UK, Summer 2019

SERVICES AND LEADERSHIP IN THE SCIENTIFIC COMMUNITY

- 2026– Co-Chair, Young Professionals Network, UC Santa Barbara — a campus network supporting postdoctoral researchers and early-career staff through professional development, networking, and community-building events. Selected to co-lead the organization's relaunch and expand its reach across campus.
- 2025– Delivered hands-on science demonstrations to elementary school students through the UCSB Materials Research Laboratory's outreach program, broadening early exposure to STEM in local schools.
- 2025– Delivered technical TOPAS tutorials on the analysis of X-ray and neutron diffraction data as part of the Diamond Light Source I11 BAG user development training program at UC Santa Barbara. Topics included Pawley and LeBail fitting, quantitative Rietveld refinement of single- and multi-phase data, structure solution methods, sequential refinements, and treatment of (an)isotropic displacement parameters.
- 2024–2026 Collaborating scientist, Partnership for Research and Education in Materials (**PREM**) — an NSF program that pairs minority-serving institutions with leading materials research centers; UCSB's PREM is a joint research and education partnership with Jackson State University, Mississippi, and the University of Texas at El Paso, Texas. Contributed experimental data to collaborative UCSB projects; delivered an invited talk and served as a poster judge at the 2025 PREM conference at Jackson State University.
- 2024–2026 Research mentor, Future Leaders in Advanced Materials (**FLAM**) — a summer research internship program sponsored by the UCSB Materials Research Laboratory and College of Engineering, hosting undergraduate students from universities across the US. Provided training in materials synthesis, characterization, data analysis, and scientific communication.
- 2022–2024 Organized and facilitated fortnightly check-ins for new and junior postgraduate researchers in the Physical Sciences Department, University of Liverpool, supporting wellbeing, mentoring, and professional development.

PROPOSALS AWARDED

- 2024 NOMAD, Spallation Neutron Source, Oak Ridge National Laboratory, US (5 days awarded total).
- 2023–2025 Beamline I11, Diamond Light Source, UK, UC Santa Barbara Batch Allocation Grant (12 days/year).
- 2023 Photon Science Institute, University of Manchester, UK (10 days awarded total).
- 2023 Beamline B18, Diamond Light Source, Didcot, UK (3 days awarded total).
- 2022 XPS: HarwellXPS, Rutherford Appleton Laboratory, Didcot, UK (2 days awarded).
- 2019–2023 Beamline I11, Diamond Light Source, UK, University of Liverpool Batch Allocation Grant (12 days/year).

PROFESSIONAL AFFILIATIONS

- 2023 Member of the American Chemical Society (ACS)
- 2022 Member of the Electrochemical Society (ECS)
- 2019 Member of the Royal Society of Chemistry (RSC)

VOLUNTEERING

- 2026– Organized and led a team of ten UCSB students and staff volunteering with United Way's Lunch Bunch summer enrichment program for underprivileged elementary schoolers, bridging the learning gap of summer vacation.
- 2022–2024 Founded a departmental book club for members of the University of Liverpool Physical Sciences Department.
- 2015–2016 Volunteered with isolated patients at Royal Liverpool University Hospital, providing companionship to those without family support.
- 2014–2015 Volunteered with Loud and Clear, a community initiative using music and performance to support mental health, wellbeing, and creative expression in Warrington, UK.
- 2013–2014 Completed National Citizen Service with the Tim Parry and Jonathan Ball Peace Foundation, developing skills in youth leadership, teamwork, peacebuilding, and community engagement in Warrington, UK.

REFERENCES**Professor Ram Seshadri**

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